Lab Report: Image classification experiments

1 Introduction

In this report we compare several models on standard benchmarks. We trained each model for 100 epochs with a batch size of 64. The dataset was split into 70 percent training data and the rest for testing.

2 Results

Figure 1 shows the f1 score during training.

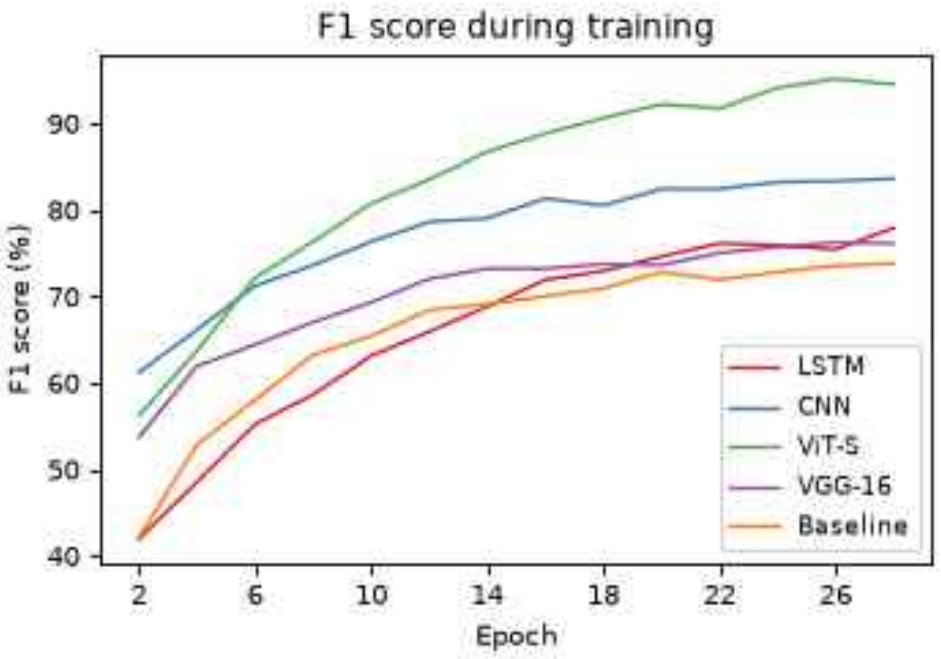


Figure 1: F1 score during training for the models in this experiment.

At epoch 2, Baseline achieves 30.4% F1 score. ViT-S outperforms VGG-16 by 1.8 percentage points at epoch 6 (Figure 1). As shown in Figure 1, CNN attains a F1 score of 53.7% at epoch 2. ViT-S attains a F1 score of 58.6% at epoch 8.

Figure 2 shows the validation accuracy by dataset.

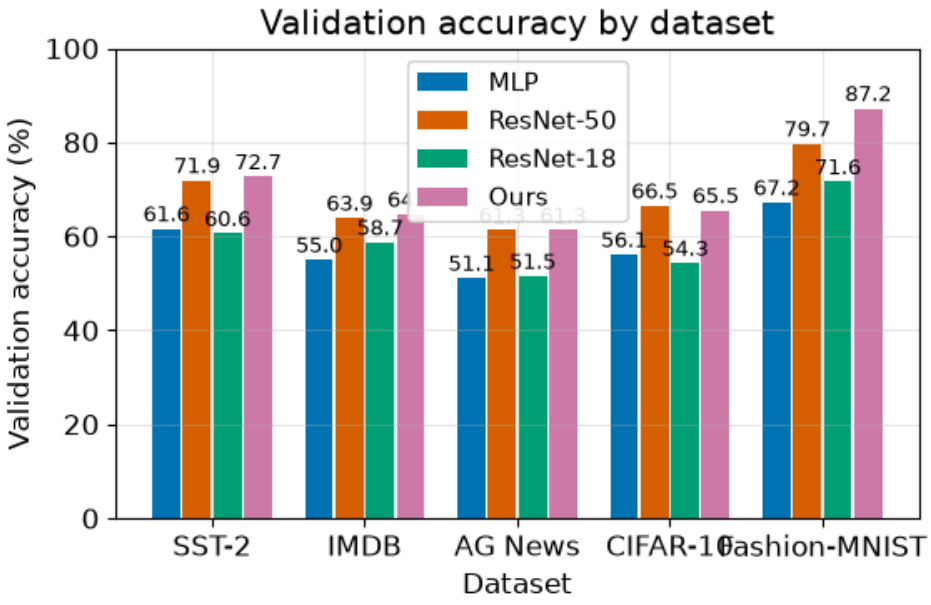


Figure 2: Validation accuracy by dataset for the models in this experiment.

Ours outperforms MLP by 31.1 percentage points on SST-2 (Figure 2). On Fashion-MNIST, ResNet-50 obtains a validation accuracy of 69.7% (Figure 2).

Figure 3 shows the validation accuracy by dataset.

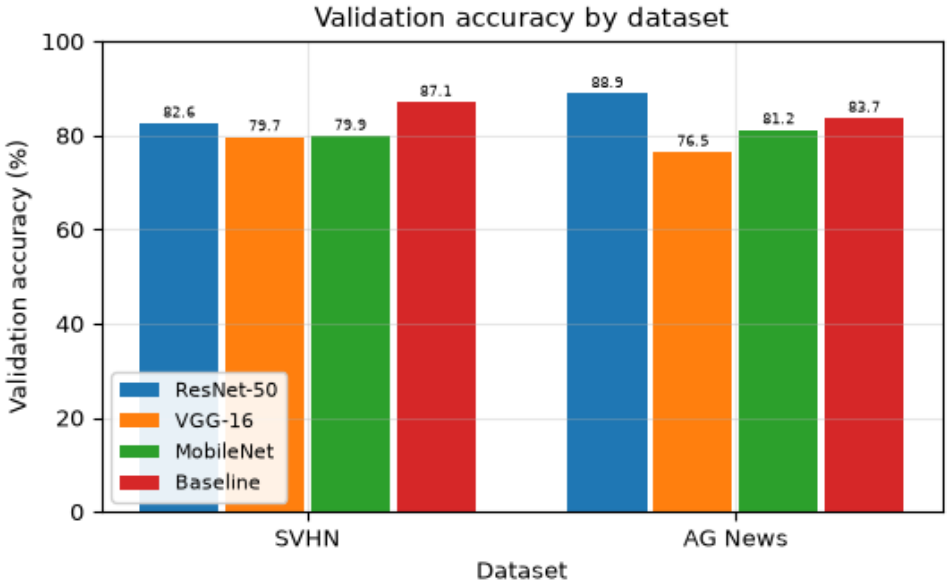


Figure 3: Validation accuracy by dataset for the models in this experiment.

VGG-16 achieves 69.7% validation accuracy on SVHN (Figure 3). As shown in Figure 3, Baseline outperforms MobileNet by 52.5 percentage points on AG News. On AG News, MobileNet obtains a validation accuracy of 131.2% (Figure 3). Baseline outperforms VGG-16 by 7 percentage points on AG News.

Figure 4 shows the f1 score by dataset.

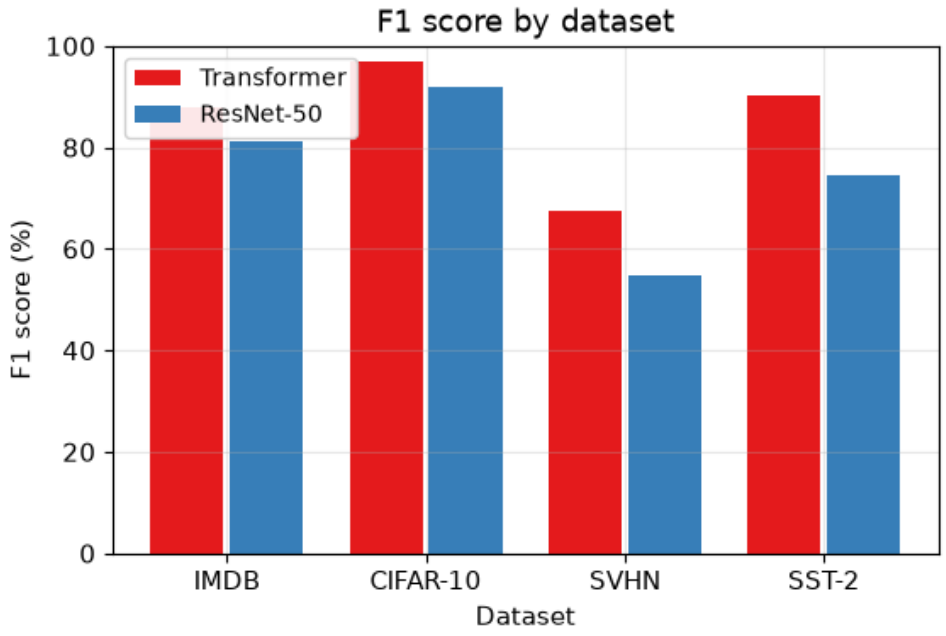


Figure 4: F1 score by dataset for the models in this experiment.

On SST-2, Transformer obtains a F1 score of 90% (Figure 4). As shown in Figure 4, on IMDB, ResNet-50 obtains a F1 score of 101.1%. On IMDB, Transformer is 6.7 points better than ResNet-50 (Figure 4). Transformer outperforms ResNet-50 by 13 percentage points on SVHN (Figure 4).

3 Discussion

On AG News, Naive Bayes obtains an accuracy of 50.7%. The differences between the models are

discussed in the next section.